MATH 201 Your name: ______________________

Learning target DF7, version 1

Compute derivatives of implicitly-defined functions.

Consider the curve defined implicitly by the equation

$$xy^3 + 2xy = 6$$

1. Use implicit differentiation to find an expression for $\frac{dy}{dx}$ that depends on x and y. Show all of your work and thinking clearly, using correct notation.

2. Notice that the point $(2,1)$ lies on the curve $xy^3 + 2xy = 6$. Find the exact slope of the tangent line to the curve at $(2,1)$. Show clearly how you determined this value, and label your work with proper notation.

MATH 201 Your name: ____________________

Learning target DF7, version 2

Compute derivatives of implicitly-defined functions.

The equation $x^3 - y^3 = 9xy$ makes a curve called the *folium of Descartes*.

1. Use implicit differentiation to find an expression for $\frac{dy}{dx}$ that depends on x and y. Show all of your work and thinking clearly, using correct notation.

2. Find the exact slope of the tangent line to this curve at the point $(-2,4)$. Show clearly how you determined this value, and label your work with proper notation.

3. (Also AD2) Find an equation for the tangent line to this curve at this point, and use your tangent line to make a reasonable guess about the y-coordinate of the point $(-2.5, \underline{\quad\quad})$.

MATH 201 Your name: ______________________________

Learning target DF7, version 3

Compute derivatives of implicitly-defined functions.

Consider the curve defined implicitly by the equation

$$x^5 - 8y^4 = \sin(y) - 7.$$

(a) Use implicit differentiation to find an expression for $\frac{dy}{dx}$ that depends on x and y. Show all of your work and thinking clearly, using correct notation.

(b) This curve contains the point $(-1.4758, 0)$ (or, pretty close to that). Find the exact slope of the tangent line to this curve at this point. Show clearly how you determined this value, and label your work with proper notation.

MATH 201 Your name: ____________________

Learning target DF7, version 4

Compute derivatives of implicitly-defined functions.

Consider the curve defined implicitly by the equation

$$\sin(x+y)+\cos(x-y)=1.$$

(a) Use implicit differentiation to find an expression for $\frac{dy}{dx}$ that depends on x and y. Show all of your work and thinking clearly, using correct notation.

(b) This curve contains the point $\left(\frac{\pi}{2},\frac{\pi}{2}\right)$. Find the exact slope of the tangent line to this curve at this point. Show clearly how you determined this value, and label your work with proper notation.